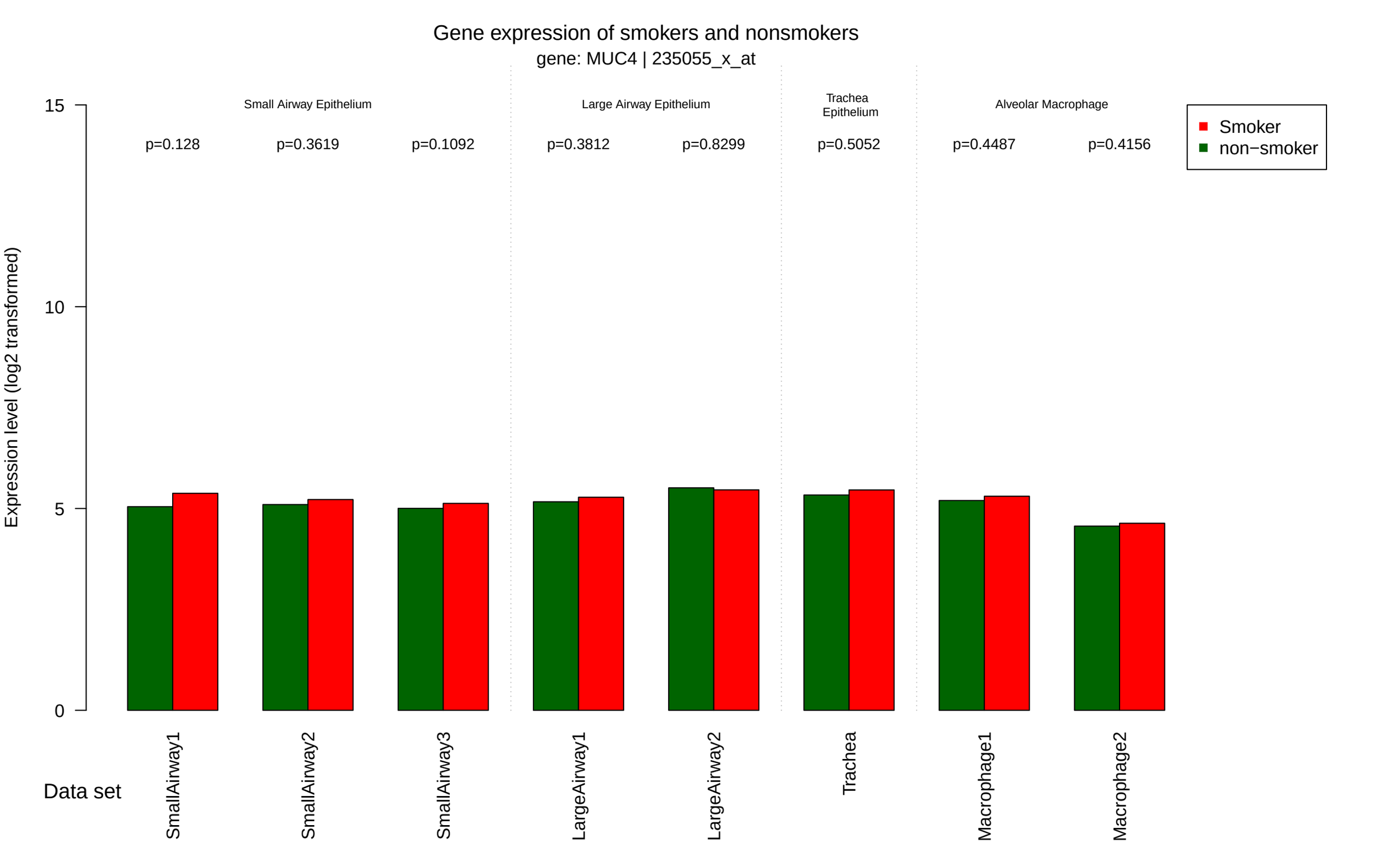
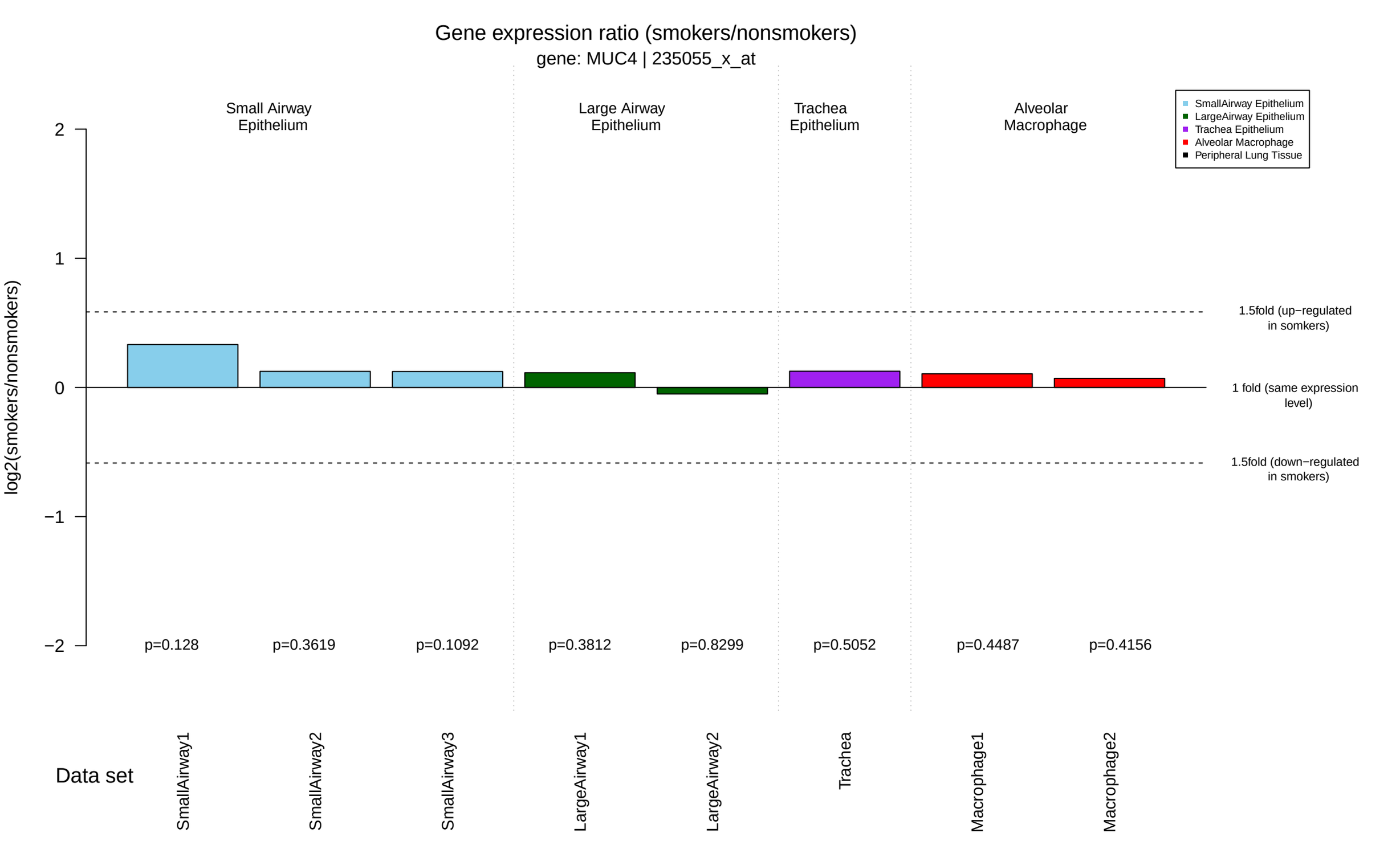
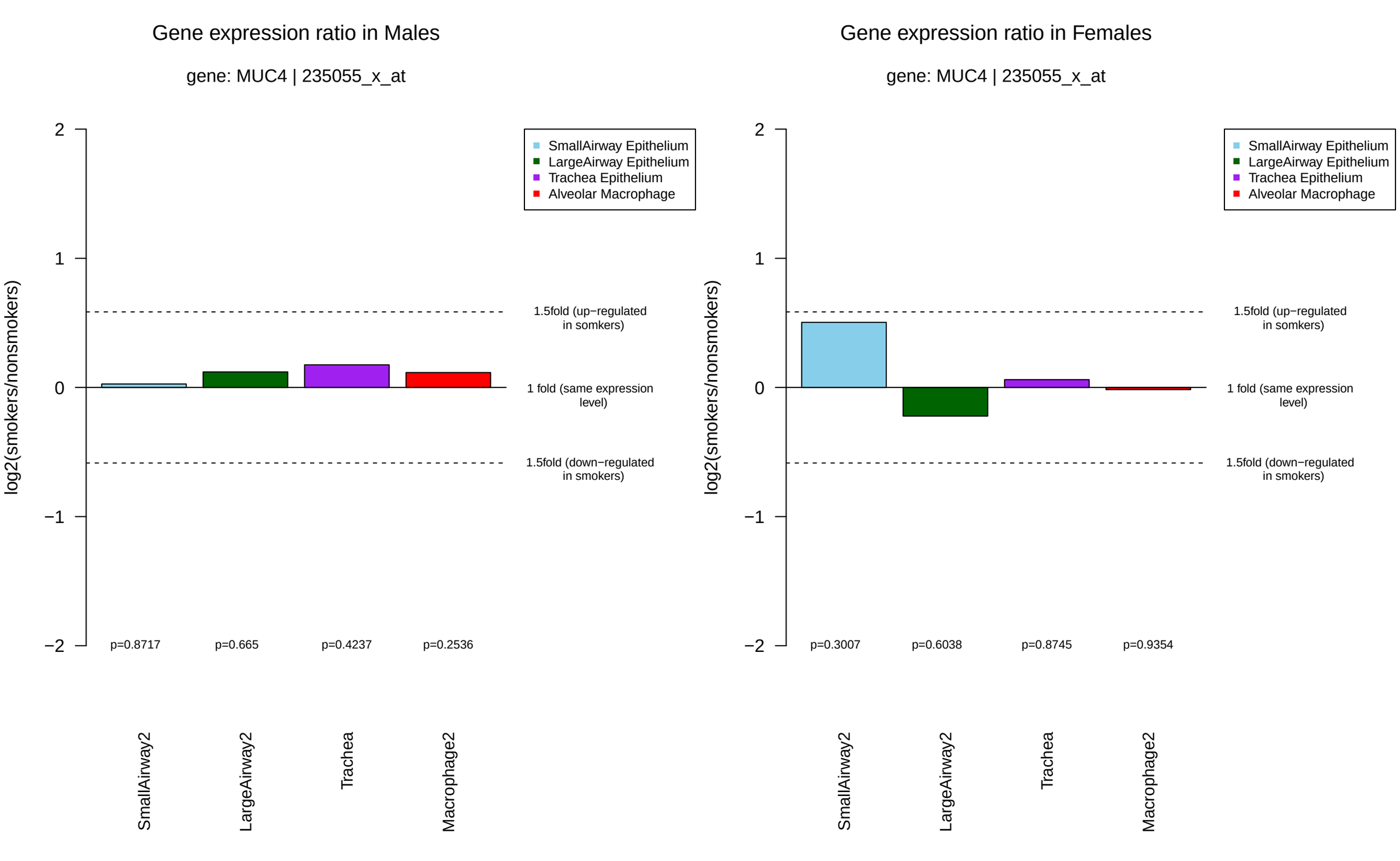
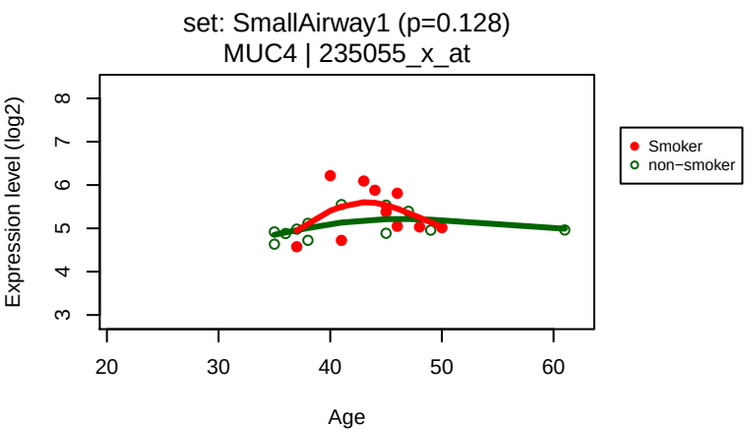
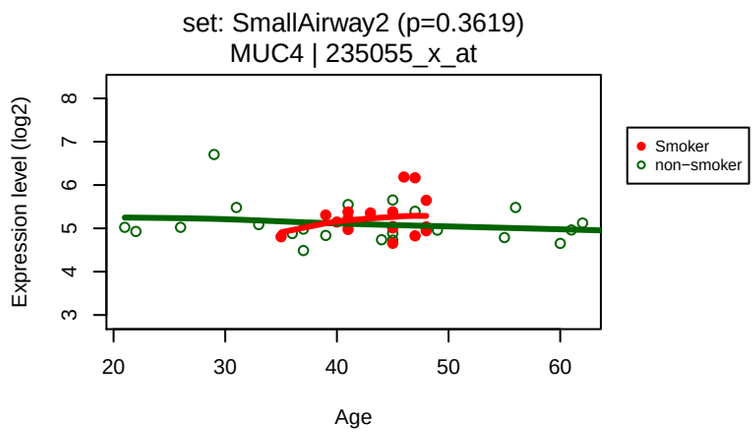
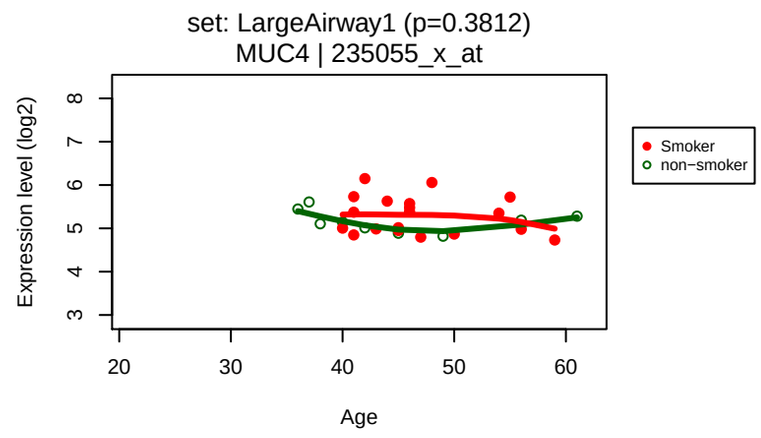

Gene expression of smokers and nonsmokers
gene: MUC4 | 235055_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.128
p=0.3619
p=0.1092
p=0.3812
p=0.8299
p=0.5052
p=0.4487
p=0.4156
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

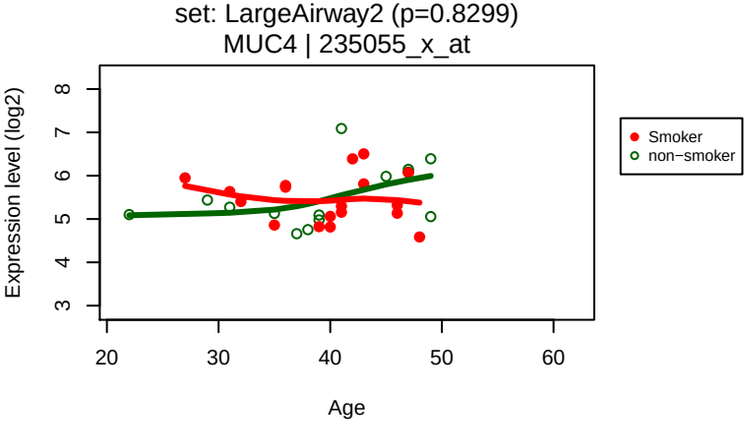

Gene expression ratio (smokers/nonsmokers)
gene: MUC4 | 235055_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.128
p=0.3619
p=0.1092
p=0.3812
p=0.8299
p=0.5052
p=0.4487
p=0.4156
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

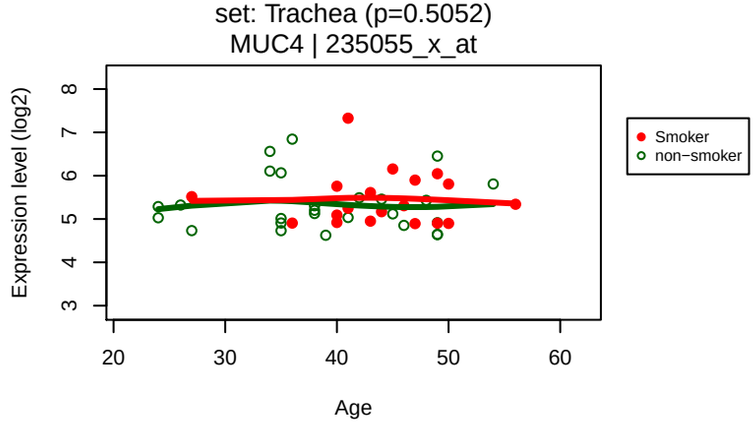

Gene expression ratio in Males
gene: MUC4 | 235055_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8717
p=0.665
p=0.4237
p=0.2536
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUC4 | 235055_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3007
p=0.6038
p=0.8745
p=0.9354
SmallAirway2
LargeAirway2
Trachea
Macrophage2

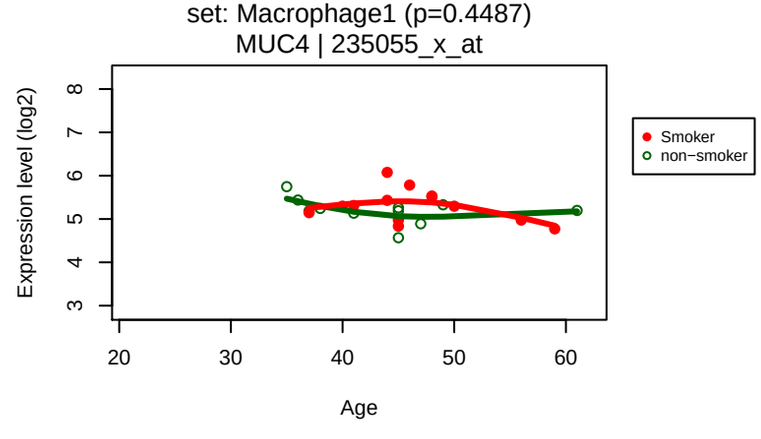

set: SmallAirway1 (p=0.128)
MUC4 | 235055_x_at
Expression level (log2)
Age
Smoker
non−smoker

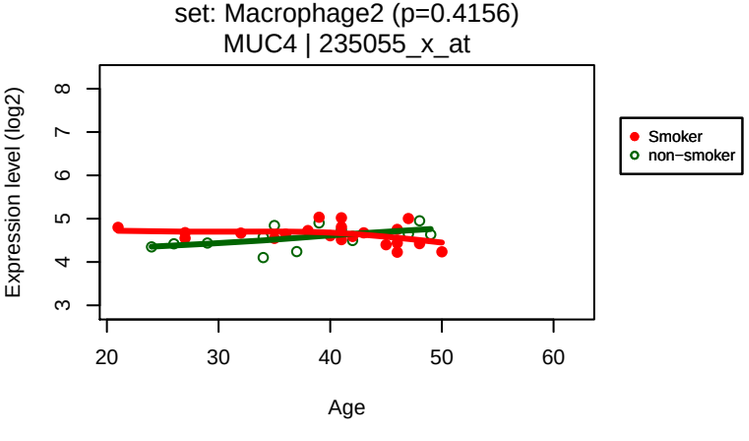

set: SmallAirway2 (p=0.3619)
MUC4 | 235055_x_at
Expression level (log2)
Age
Smoker
non−smoker

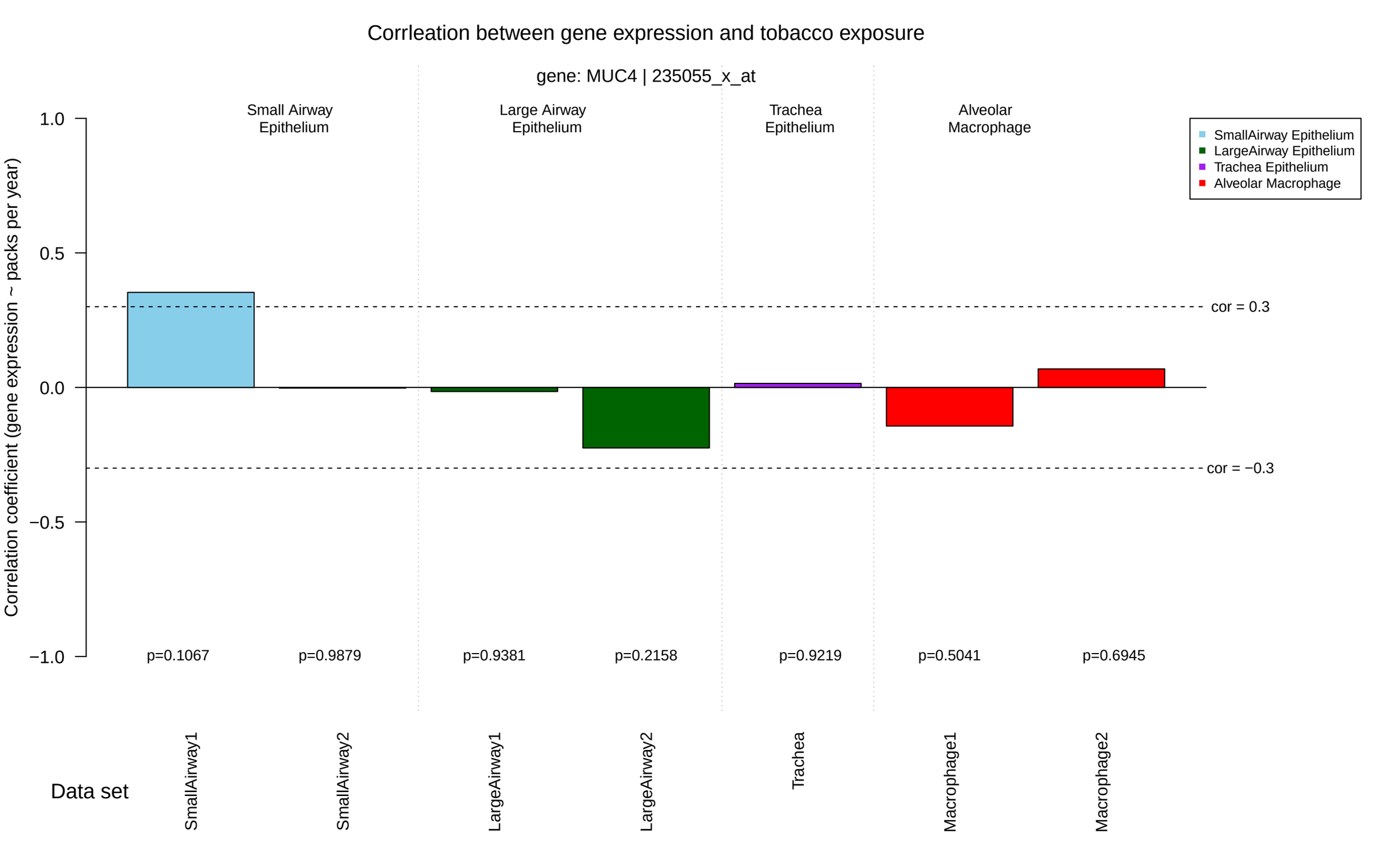

set: LargeAirway1 (p=0.3812)
MUC4 | 235055_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8299)
MUC4 | 235055_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5052)
MUC4 | 235055_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4487)
MUC4 | 235055_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4156)
MUC4 | 235055_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC4 | 235055_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1067
p=0.9879
p=0.9381
p=0.2158
p=0.9219
p=0.5041
p=0.6945
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2